LeetCode Problem Set for DSA Preparation

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1. Linear Search Problems

Beginner

1. Two Sum
2. Find Numbers with Even Number of Digits
3. Richest Customer Wealth
4. Maximum Number of Words Found in Sentences
5. Check if Array Is Sorted and Rotated
6. Linear Search in Array
7. Find the Highest Altitude
8. Kids With the Greatest Number of Candies

Intermediate

9. Search Insert Position
10. Find Peak Element
11. Majority Element
12. Missing Number
13. Single Number
14. Find All Numbers Disappeared in an Array
15. Longest Consecutive Sequence

Advanced

16. First Missing Positive
17. Kth Largest Element in an Array
18. Top K Frequent Elements

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2. Binary Search Problems

Basic Binary Search

1. Binary Search
2. Search Insert Position
3. Guess Number Higher or Lower
4. Sqrt(x)
5. Valid Perfect Square

Binary Search on Arrays

6. Find First and Last Position of Element in Sorted Array
7. Search in Rotated Sorted Array
8. Search in Rotated Sorted Array II
9. Find Minimum in Rotated Sorted Array
10. Peak Index in a Mountain Array
11. Find Peak Element

Binary Search on Answers

12. Koko Eating Bananas
13. Capacity To Ship Packages Within D Days
14. Split Array Largest Sum
15. Minimized Maximum of Products Distributed to Any Store
16. Aggressive Cows (important outside LeetCode too)

Advanced

17. Median of Two Sorted Arrays
18. Find K Closest Elements
19. Time Based Key-Value Store
20. Search a 2D Matrix
21. Search a 2D Matrix II

---

3. Stack Problems

Stack Basics

1. Valid Parentheses
2. Implement Stack using Queues
3. Baseball Game
4. Backspace String Compare
5. Remove Outermost Parentheses

Monotonic Stack

6. Next Greater Element I
7. Next Greater Element II
8. Daily Temperatures
9. Stock Span Problem
10. Largest Rectangle in Histogram

Expression Problems

11. Evaluate Reverse Polish Notation
12. Basic Calculator
13. Basic Calculator II

Advanced Stack

14. Min Stack
15. Decode String
16. Asteroid Collision
17. Remove K Digits
18. Trapping Rain Water

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4. Queue Problems

Queue Basics

1. Implement Queue using Stacks
2. Number of Recent Calls
3. Moving Average from Data Stream
4. Time Needed to Buy Tickets

Circular Queue

5. Design Circular Queue
6. Design Circular Deque

Priority Queue / Heap

7. Kth Largest Element in a Stream
8. Last Stone Weight
9. Top K Frequent Elements
10. Merge K Sorted Lists
11. Find Median from Data Stream

BFS Using Queue

12. Binary Tree Level Order Traversal
13. Rotting Oranges
14. 01 Matrix
15. Word Ladder

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5. Linked List Problems

A. Singly Linked List

Beginner

1. Reverse Linked List
2. Middle of the Linked List

3. Linked List Cycle
4. Remove Linked List Elements
5. Delete Node in a Linked List
6. Merge Two Sorted Lists

Intermediate

7. Remove Nth Node From End of List
8. Palindrome Linked List
9. Intersection of Two Linked Lists
10. Reorder List
11. Odd Even Linked List
12. Add Two Numbers

Advanced

13. Reverse Nodes in k-Group
14. Copy List with Random Pointer
15. LRU Cache
16. Flatten a Multilevel Doubly Linked List

B. Doubly Linked List

1. Design Browser History
2. Design Linked List
3. Flatten a Multilevel Doubly Linked List
4. LRU Cache

C. Circular Linked List

(LeetCode has fewer direct circular linked list problems, but these help)

1. Design Circular Queue
2. Design Circular Deque
3. Josephus Problem (outside LeetCode important)
4. Insert into a Sorted Circular Linked List

D. Doubly Circular Linked List

Mostly implementation-based and asked in interviews rather than direct LeetCode problems.

Practice:

1. Browser Navigation Systems
2. Music Playlist Simulation
3. Round Robin Scheduling
4. Custom Cache Implementations

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6. Two Pointer Problems

Beginner

1. Valid Palindrome
2. Two Sum II - Input Array Is Sorted
3. Reverse String
4. Move Zeroes
5. Remove Duplicates from Sorted Array

Intermediate

6. 3Sum
7. Container With Most Water
8. Sort Colors
9. Squares of a Sorted Array
10. Partition Labels

Advanced

11. Trapping Rain Water
12. 4Sum
13. Minimum Window Subsequence

14. Boats to Save People

---

7. Sliding Window Problems

Fixed Size Window

1. Maximum Average Subarray I
2. Contains Duplicate II
3. Maximum Number of Vowels in a Substring of Given Length

Variable Size Window

4. Longest Substring Without Repeating Characters
5. Minimum Size Subarray Sum
6. Fruit Into Baskets
7. Permutation in String
8. Longest Repeating Character Replacement
9. Minimum Window Substring

Advanced Sliding Window

10. Sliding Window Maximum
11. Subarrays with K Different Integers
12. Count Number of Nice Subarrays
13. Binary Subarrays With Sum

---

8. Sorting Algorithm Problems

Bubble Sort Practice

1. Sort Array By Parity

2. Height Checker
3. Relative Sort Array

Selection Sort Practice

4. Minimum Difference Between Highest and Lowest of K Scores
5. Array Partition

Insertion Sort Practice

6. Insertion Sort List
7. Sort List

Merge Sort Problems

8. Sort an Array
9. Count of Smaller Numbers After Self
10. Reverse Pairs

Quick Sort Problems

11. Kth Largest Element in an Array
12. Sort Colors
13. Top K Frequent Elements

Heap Sort / Priority Queue

14. Merge K Sorted Lists
15. Find K Pairs with Smallest Sums
16. Task Scheduler

Counting Sort / Radix Sort

17. Maximum Gap
18. Sort Characters By Frequency
19. Relative Sort Array

Bucket Sort

20. Top K Frequent Elements
21. Contains Duplicate III

Best Order to Solve

Phase 1 — Basics

- Linear Search
- Basic Sorting
- Stack Basics
- Queue Basics
- Singly Linked List Basics

Phase 2 — Intermediate

- Binary Search
- Two Pointers
- Sliding Window
- Monotonic Stack
- Circular Queue

Phase 3 — Advanced

- Heap / Priority Queue

- Merge Sort / Quick Sort
- Advanced Linked Lists
- Advanced Sliding Window
- Binary Search on Answers

Most Important Interview Problems

If you have limited time, prioritize these:

MUST DO

1. Two Sum
2. Binary Search
3. Valid Parentheses
4. Reverse Linked List
5. Linked List Cycle
6. Merge Two Sorted Lists
7. Longest Substring Without Repeating Characters
8. Container With Most Water
9. 3Sum
10. Sliding Window Maximum
11. Daily Temperatures
12. Koko Eating Bananas
13. LRU Cache
14. Merge K Sorted Lists
15. Trapping Rain Water

Recommended Practice Strategy

Daily Routine

- 2 Easy Problems
- 2 Medium Problems
- 1 Revision Problem

Weekly Focus

Week	Topic
1	Arrays + Linear Search
2	Binary Search
3	Stack + Queue
4	Linked List
5	Two Pointers
6	Sliding Window
7	Sorting
8	Mixed Revision

Best Problems for Building Logic

Best for Thinking

- 3Sum
- Trapping Rain Water
- Sliding Window Maximum
- Median of Two Sorted Arrays
- Merge K Sorted Lists
- Minimum Window Substring

Best Problems for Interviews

FAANG Frequently Asked

- Valid Parentheses
- LRU Cache
- Koko Eating Bananas
- Binary Search
- Reverse Linked List
- Merge Intervals
- Top K Frequent Elements
- Daily Temperatures

- Sliding Window Maximum
- Word Ladder

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1. Trees (Binary Tree + BST)

Basics

1. Maximum Depth of Binary Tree
2. Same Tree
3. Invert Binary Tree
4. Symmetric Tree
5. Subtree of Another Tree

Traversals (DFS)

6. Binary Tree Inorder Traversal
7. Binary Tree Preorder Traversal
8. Binary Tree Postorder Traversal
9. Binary Tree Level Order Traversal

Intermediate

10. Diameter of Binary Tree
11. Balanced Binary Tree
12. Path Sum
13. Path Sum II
14. Lowest Common Ancestor of a Binary Tree
15. Validate Binary Search Tree
16. Kth Smallest Element in a BST

Advanced

17. Binary Tree Maximum Path Sum
18. Serialize and Deserialize Binary Tree

19. Construct Binary Tree from Preorder and Inorder Traversal
20. Flatten Binary Tree to Linked List
21. Recover Binary Search Tree

---

2. Graphs

Basics

1. Find if Path Exists in Graph
2. Number of Connected Components
3. Clone Graph

DFS Based

4. Number of Islands
5. Max Area of Island
6. Surrounded Regions
7. Keys and Rooms

BFS Based

8. Rotting Oranges
9. 01 Matrix
10. Word Ladder
11. Shortest Path in Binary Matrix

Topological Sort

12. Course Schedule
13. Course Schedule II
14. Alien Dictionary

Shortest Path

15. Network Delay Time
16. Cheapest Flights Within K Stops
17. Dijkstra Algorithm (important concept)

Advanced Graph

18. Redundant Connection
19. Critical Connections in a Network
20. Reconstruct Itinerary
21. Minimum Spanning Tree (Prim's / Kruskal)

---

3. BFS & DFS (Combined Practice)

Must Do

1. Number of Islands
2. Flood Fill
3. Rotting Oranges
4. Clone Graph
5. Word Ladder

Tree + Graph DFS

6. Path Sum
7. Binary Tree Paths
8. All Paths From Source to Target

Advanced

9. Pacific Atlantic Water Flow
10. Course Schedule (Cycle Detection)
11. Detect Cycle in Graph

12. Shortest Path in Binary Matrix

---

4. Dynamic Programming (DP)

DP Basics (1D)

1. Climbing Stairs
2. House Robber
3. Fibonacci Number
4. Min Cost Climbing Stairs

2D DP

5. Unique Paths
6. Unique Paths II
7. Minimum Path Sum
8. Dungeon Game

Subsequence / Subset DP

9. Longest Increasing Subsequence
10. Longest Common Subsequence
11. Partition Equal Subset Sum
12. Target Sum

String DP

13. Longest Palindromic Substring
14. Longest Palindromic Subsequence
15. Edit Distance
16. Distinct Subsequences

Knapsack Pattern

17. 0/1 Knapsack (classic)
18. Coin Change
19. Coin Change II
20. Combination Sum IV

Advanced DP

21. Burst Balloons
22. Matrix Chain Multiplication
23. Regular Expression Matching
24. Wildcard Matching

---

5. Greedy Algorithms

Basics

1. Assign Cookies
2. Best Time to Buy and Sell Stock
3. Maximum Subarray (Kadane's Algorithm)

Interval Problems

4. Merge Intervals
5. Non-overlapping Intervals
6. Insert Interval
7. Meeting Rooms

Advanced Greedy

8. Jump Game
9. Jump Game II
10. Gas Station

11. Task Scheduler

Hard Greedy

12. Candy
13. Minimum Number of Arrows to Burst Balloons
14. Partition Labels

---

6. Backtracking

Basics

1. Subsets
2. Permutations
3. Combinations

Intermediate

4. Combination Sum
5. Combination Sum II
6. Letter Combinations of a Phone Number

Advanced

7. N-Queens
8. Sudoku Solver
9. Word Search
10. Palindrome Partitioning

---

7. Advanced Data Structures

Heap / Priority Queue

1. Kth Largest Element
2. Top K Frequent Elements
3. Find Median from Data Stream

Trie

1. Implement Trie
2. Word Search II
3. Replace Words

Union Find (Disjoint Set)

1. Number of Provinces
2. Redundant Connection
3. Accounts Merge
4. Number of Islands II

Segment Tree / Fenwick Tree

1. Range Sum Query
2. Range Sum Query Mutable
3. Count of Smaller Numbers After Self

---

8. Bit Manipulation

Basics

1. Single Number
2. Missing Number
3. Power of Two

Intermediate

4. Counting Bits
5. Sum of Two Integers
6. Subsets

Advanced

7. Maximum XOR of Two Numbers in an Array
8. Bitwise AND of Numbers Range

---

9. Hard Core Interview Problems

Must Do

1. Median of Two Sorted Arrays
2. Merge K Sorted Lists
3. LRU Cache
4. Sliding Window Maximum
5. Trapping Rain Water
6. Word Ladder
7. Serialize and Deserialize Binary Tree
8. Longest Increasing Subsequence
9. Edit Distance
10. Burst Balloons

Best Learning Path

Phase 1

- Trees Basics
- DFS / BFS
- Easy DP

Phase 2

- Graphs
- Medium DP
- Greedy

Phase 3

- Advanced DP
- Union Find
- Trie
- Backtracking

Phase 4

- Hard Problems
- Mixed Practice

If You Want Maximum Results

Follow this:

Daily:

- 2 Easy
- 2 Medium
- 1 Hard (optional)

Weekly Focus:

- Week 1 → Trees
- Week 2 → Graphs
- Week 3 → DP
- Week 4 → Mixed + Revision

MUST DO PROBLEMS IN EACH TOPIC

1. Arrays

These are asked in almost every company.

Problem	Difficulty
Two Sum	Easy
Best Time to Buy and Sell Stock	Easy
Maximum Subarray (Kadane’s Algorithm)	Medium
Contains Duplicate	Easy
Product of Array Except Self	Medium
Move Zeroes	Easy
Merge Sorted Arrays	Easy
Rotate Array	Medium
Find Missing Number	Easy
Majority Element	Easy
Trapping Rain Water	Hard
Container With Most Water	Medium
Spiral Matrix	Medium
Set Matrix Zeroes	Medium
Next Permutation	Medium

---

2. Strings

Very common in service-based companies and OA rounds.

Problem	Difficulty
Valid Palindrome	Easy

Reverse String	Easy
Longest Common Prefix	Easy
Valid Anagram	Easy
Group Anagrams	Medium
Longest Substring Without Repeating Characters	Medium
String Compression	Medium
Count and Say	Medium
Roman to Integer	Easy
Integer to Roman	Medium
Minimum Window Substring	Hard
Longest Palindromic Substring	Medium
Implement strstr()	Easy
Decode String	Medium

---

3. Sliding Window Problems

VERY frequently asked in product companies.

Problem	Difficulty
Maximum Sum Subarray of Size K	Easy
Longest Substring Without Repeating Characters	Medium
Minimum Size Subarray Sum	Medium
Fruits Into Baskets	Medium
Permutation in String	Medium
Sliding Window Maximum	Hard

---

4. HashMap / Hashing Problems

Problem	Difficulty
Two Sum	Easy
Top K Frequent Elements	Medium
Subarray Sum Equals K	Medium
Happy Number	Easy
Longest Consecutive Sequence	Medium
Find Duplicate Number	Medium
First Unique Character	Easy

---

5. Linked List Problems

VERY important for interviews.

Problem	Difficulty
Reverse Linked List	Easy
Detect Cycle in Linked List	Easy
Merge Two Sorted Lists	Easy
Middle of Linked List	Easy
Remove Nth Node From End	Medium
Palindrome Linked List	Easy
Intersection of Two Linked Lists	Easy
Copy List with Random Pointer	Hard
LRU Cache	Hard

---

6. Stack & Queue Problems

Problem	Difficulty
Valid Parentheses	Easy
Min Stack	Medium
Implement Queue using Stacks	Easy
Daily Temperatures	Medium
Next Greater Element	Easy
Largest Rectangle in Histogram	Hard
Evaluate Reverse Polish Notation	Medium

---

7. Binary Search Problems

Asked extremely often in Amazon, Microsoft, Google.

Problem	Difficulty
Binary Search	Easy
Search Insert Position	Easy
First and Last Position in Sorted Array	Medium
Search in Rotated Sorted Array	Medium
Find Peak Element	Medium
Median of Two Sorted Arrays	Hard
Koko Eating Bananas	Medium

---

8. Recursion & Backtracking

Problem	Difficulty
Subsets	Medium
Permutations	Medium

Combination Sum	Medium
Generate Parentheses	Medium
N Queens	Hard
Sudoku Solver	Hard
Letter Combinations of Phone Number	Medium

9. Trees & Binary Trees

MOST IMPORTANT for product companies.

Problem	Difficulty
Maximum Depth of Binary Tree	Easy
Inorder Traversal	Easy
Level Order Traversal	Medium
Diameter of Binary Tree	Easy
Validate BST	Medium
Lowest Common Ancestor	Medium
Symmetric Tree	Easy
Path Sum	Easy
Serialize and Deserialize Binary Tree	Hard

10. Binary Search Trees (BST)

Problem	Difficulty
Search in BST	Easy
Insert into BST	Medium
Delete Node in BST	Medium

Kth Smallest Element in BST	Medium
Convert Sorted Array to BST	Easy

---

11. Heap / Priority Queue

Problem	Difficulty
Kth Largest Element	Medium
Top K Frequent Elements	Medium
Merge K Sorted Lists	Hard
Find Median from Data Stream	Hard
Task Scheduler	Medium

---

12. Graph Problems

VERY important for top product companies.

Problem	Difficulty
Number of Islands	Medium
Clone Graph	Medium
Course Schedule	Medium
BFS Traversal	Easy
DFS Traversal	Easy
Rotten Oranges	Medium
Word Ladder	Hard
Dijkstra Algorithm	Hard
Flood Fill	Easy

---

13. Dynamic Programming (DP)

Most feared but highly asked.

Problem	Difficulty
Climbing Stairs	Easy
House Robber	Medium
Coin Change	Medium
Longest Increasing Subsequence	Medium
Longest Common Subsequence	Medium
0/1 Knapsack	Medium
Partition Equal Subset Sum	Medium
Edit Distance	Hard
DP Fibonacci	Easy

---

14. Greedy Algorithms

Problem	Difficulty
Jump Game	Medium
Gas Station	Medium
Assign Cookies	Easy
Non Overlapping Intervals	Medium
Activity Selection	Medium

---

15. Bit Manipulation

Frequently asked in OA rounds.

Problem	Difficulty
Single Number	Easy
Counting Bits	Easy
Number of 1 Bits	Easy
Missing Number	Easy
Reverse Bits	Easy

---

16. SQL Questions (VERY COMMON)

Service-based companies ask these heavily.

Topic	Important Concepts
Joins	INNER, LEFT, RIGHT
Group By	Aggregation
Window Functions	RANK, DENSE_RANK
Subqueries	Nested queries
CTE	WITH clause
Indexes	Optimization
Normalization	DBMS
Stored Procedures	Advanced SQL

Common SQL Problems:

- Second Highest Salary
- Employees Earning More Than Managers
- Duplicate Emails
- Top 3 Salaries by Department
- Monthly Revenue Report

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17. System Design (Experienced Roles)

Asked in companies like:

- Google
- Amazon
- Netflix
- Uber

Topics:

- URL Shortener
- Chat Application
- Notification System
- Rate Limiter
- Load Balancer
- Caching
- Database Scaling

TOP 25 MUST-DO PROBLEMS

If you prepare only these thoroughly, you can clear many interviews.

1. Two Sum
2. Valid Parentheses
3. Reverse Linked List
4. Maximum Subarray
5. Binary Search
6. Merge Intervals
7. Longest Substring Without Repeating Characters
8. Product of Array Except Self
9. Best Time to Buy and Sell Stock
10. Group Anagrams
11. Number of Islands
12. Clone Graph
13. Course Schedule
14. Kth Largest Element
15. Climbing Stairs
16. House Robber
17. Coin Change
18. Longest Increasing Subsequence
19. LRU Cache
20. Sliding Window Maximum
21. Merge K Sorted Lists
22. Search in Rotated Sorted Array
23. Top K Frequent Elements
24. Combination Sum

25. Lowest Common Ancestor

Company-Wise Focus

Company Type	Focus Areas
Product-Based	DSA, Graphs, DP, Trees, System Design
Service-Based	Arrays, Strings, SQL, OOPs
Startups	Practical Coding + APIs + DB
FAANG	Hard DSA + Optimization

Best Platforms

- LeetCode
- GeeksforGeeks
- HackerRank
- CodeStudio (Coding Ninjas)
- InterviewBit

Recommended Preparation Order

1. Arrays
2. Strings
3. Hashing
4. Sliding Window
5. Linked Lists
6. Stack & Queue
7. Binary Search
8. Trees
9. Heap
10. Graphs
11. Recursion
12. Dynamic Programming
13. System Design

---

1. STACK (Basic Stack Problems)

Concepts Covered

- Push / Pop
- LIFO
- Stack simulation
- Expression evaluation
- Balanced symbols

What Companies Test

Can you think in **LIFO order** and simulate operations efficiently?

Beginner Questions

1. Valid Parentheses

Concept:
Basic stack push-pop.

Focus:

opening bracket → push
closing bracket → pop & validate

---

2. Implement Stack using Queues

Concept:

Data structure implementation.

---

3. Baseball Game

Concept:

Stack simulation.

---

4. Backspace String Compare

Concept:

String stack simulation.

Intermediate Questions

5. Evaluate Reverse Polish Notation

Concept:

Postfix expression evaluation.

Important for:

compiler concepts
expression parsing

---

6. Decode String

Concept:

Nested stack processing.

---

7. Simplify Path

Concept:

Real-world stack simulation.

Hard

8. Basic Calculator

Concept:

Expression parsing.

---

2. MIN STACK

Concepts Covered

- Auxiliary stack
- O(1) minimum retrieval
- Design problems

What Companies Test

Can you optimize:

get minimum in constant time

Must Solve

1. Min Stack

Core problem.

---

2. Max Stack

Variation of min stack.

---

3. Design Browser History

Uses stack logic.

---

4. Design a Stack With Increment Operation

Design-based stack question.

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3. MONOTONIC STACK

Concepts Covered

- Next Greater
- Next Smaller
- Previous Greater
- Previous Smaller
- Nearest element problems

What Companies Test

Can you reduce:

$O(n^2)$
→
$O(n)$

using stack optimization?

Stage 1 — Easy

1. Next Greater Element I

Pattern:

Next Greater.

---

2. Daily Temperatures

Pattern:

Nearest greater future value.

---

3. Final Prices With a Special Discount

Pattern:

Next smaller element.

---

4. Stock Span Problem

Pattern:

Previous greater.

Very important.

Stage 2 — Medium

5. Online Stock Span

Dynamic monotonic stack.

---

6. Remove K Digits

Greedy + Monotonic Stack.

---

7. 132 Pattern

Pattern recognition.

---

8. Car Fleet

Stack-based optimization.

Stage 3 — Hard

9. Largest Rectangle in Histogram

Most important.

---

10. Maximal Rectangle

Histogram extension.

---

11. Trapping Rain Water

Stack + two pointer.

---

12. Sum of Subarray Minimums

Extremely important interview problem.

---

13. Shortest Unsorted Continuous Subarray

Advanced monotonic usage.

---

4. QUEUE

Concepts Covered

- FIFO
- Simulation
- BFS
- Scheduling

What Companies Test

Understanding queue-based processing.

Beginner

1. Implement Queue using Stacks

---

2. Number of Recent Calls

Queue sliding window.

---

3. Moving Average from Data Stream

Queue maintenance.

---

4. Time Needed to Buy Tickets

Simulation.

Medium

5. Dota2 Senate

Queue strategy.

---

6. Reveal Cards in Increasing Order

Queue simulation.

---

7. Rotting Oranges

Multi-source BFS.

Very important.

---

8. Open the Lock

Classic BFS queue problem.

---

5. DEQUE (Double Ended Queue)

Concepts Covered

Push/Pop from both ends.

Very important for:

sliding window

monotonic queue

Must Solve

1. Sliding Window Maximum

Most important deque question.

---

2. Shortest Subarray with Sum at Least K

Advanced monotonic deque.

---

3. Jump Game VI

Deque optimization.

---

6. PRIORITY QUEUE

Concepts Covered

- Highest priority first
- Dynamic ordering

What Companies Test

Can you manage data dynamically?

Beginner

1. Last Stone Weight

Heap basics.

---

2. Kth Largest Element in a Stream

Streaming heap.

Medium

3. Top K Frequent Elements

Very important.

---

4. K Closest Points to Origin

Heap pattern.

---

5. Task Scheduler

Heap + greedy.

---

6. Find K Pairs with Smallest Sums

Advanced heap.

---

7. HEAP (Min Heap + Max Heap)

Concepts Covered

- Insert
- Delete
- Top-K
- Running median

Keywords to Identify

If question says:

Top K

largest
smallest
priority
stream
frequent
closest

Think:

Heap

Beginner

1. Kth Largest Element in an Array

Must solve.

---

2. Sort Characters By Frequency

Frequency heap.

---

3. Relative Ranks

Sorting + heap.

Medium

4. Merge K Sorted Lists

Very important.

---

5. Ugly Number II

Heap generation.

---

6. Reorganize String

Max heap.

---

7. Furthest Building You Can Reach

Greedy + heap.

Hard

8. Find Median from Data Stream

Top interview problem.

Two heap concept.

---

9. IPO

Greedy + heap.

---

10. Sliding Window Median

Advanced heap.

---

8. SLIDING WINDOW MAXIMUM

Concept

Monotonic deque.

Questions

1. Sliding Window Maximum
2. Longest Continuous Subarray With Absolute Diff Less Than or Equal to Limit
3. Constrained Subsequence Sum

---

9. HISTOGRAM / RECTANGLE PROBLEMS

Pattern

Nearest smaller left/right.

Questions

1. Largest Rectangle in Histogram
2. Maximal Rectangle
3. Sum of Subarray Minimums
4. Maximum Width Ramp

Company-Wise Priority

TCS / Infosys / Cognizant

Focus:

Stack
Queue
Heap Basics
Valid Parentheses
Kth Largest

Deloitte / Capgemini / Accenture

Focus:

Monotonic Stack
Heap
Sliding Window Maximum
Top K Frequent

Amazon / Microsoft / Walmart

Focus:

Largest Rectangle
Median Stream
Monotonic Stack
Deque
Greedy + Heap

Google / Uber

Focus:

Hard optimization
Heap + Graph
Advanced monotonic stack
Streaming problems

---

1. Basic Hashing / Frequency Count Problems

(**Learn counting, frequency maps, duplicates**)

These teach:

- Frequency counting
- Existence checking
- Fast lookup (`O(1)`)
- Count maps

Easy

1. Contains Duplicate (**217**)
   Concept: HashSet for duplicates
2. Valid Anagram (**242**)
   Concept: Frequency HashMap
3. Intersection of Two Arrays (**349**)
   Concept: HashSet
4. Intersection of Two Arrays II (**350**)
   Concept: Frequency map

5. Unique Number of Occurrences (**1207**)
   Concept: Count + HashSet
6. Find the Difference (**389**)
   Concept: Character counting
7. Single Number (**136**)
   Concept: HashSet / XOR
8. Happy Number (**202**)
   Concept: HashSet cycle detection

---

2. HashMap Lookup Problems

(**Find complement, previous value lookup**)

These are **VERY IMPORTANT for interviews**

Easy → Medium

1. Two Sum (**1**)
   Most important HashMap problem

Keyword:

> "Find pair"

Pattern:

if target - num in hashmap

2. Contains Duplicate II (**219**)
   Concept: Value + index tracking
3. Isomorphic Strings (**205**)
   Concept: Character mapping
4. Word Pattern (**290**)
   Concept: Bi-directional HashMap
5. Find All Numbers Disappeared in an Array (**448**)
   Concept: Seen numbers
6. Majority Element (**169**)
   Concept: Frequency HashMap
7. Top K Frequent Elements (**347**)
   Concept: Frequency HashMap + Heap
8. Sort Characters By Frequency (**451**)
   Concept: Frequency counting

---

3. Prefix Sum + HashMap Problems

(**Very high interview frequency**)

Used when:

“Subarray sum = k”

Medium

1. Subarray Sum Equals K (**560**)

Must Learn

Pattern:

prefix_sum - k

2. Continuous Subarray Sum (**523**)
3. Contiguous Array (**525**)
4. Maximum Size Subarray Sum Equals k (**325**)
5. Binary Subarrays With Sum (**930**)
6. Count Number of Nice Subarrays (**1248**)

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4. String Hashing Problems

(**Frequency maps in strings**)

Medium

1. Group Anagrams (**49**)

Very important

Concept:

tuple(count)

or sorting key

2. Longest Palindrome (**409**)
3. First Unique Character in a String (**387**)
4. Ransom Note (**383**)

5. Longest Substring Without Repeating Characters (**3**)

HashSet + Sliding Window

6. Minimum Window Substring (**76**)

Advanced HashMap

7. Find All Anagrams in a String (**438**)

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5. HashSet Problems

(**Uniqueness / duplicate detection**)

Easy → Medium

1. Longest Consecutive Sequence (**128**)

Top Interview Problem

Pattern:

if num - 1 not in set

2. Missing Number (**268**)
3. Find Duplicate Number (**287**)
4. Jewels and Stones (**771**)
5. Check if N and Its Double Exist (**1346**)

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6. Advanced HashMap Problems

(**Frequently asked in product companies**)

Medium → Hard

1. LRU Cache (**146**)

HashMap + Doubly Linked List

Very famous interview question.

2. Insert Delete GetRandom O(1) (**380**)

HashMap + Array

3. Design HashMap (**706**)
4. Design HashSet (**705**)
5. Time Based Key Value Store (**981**)
6. Snapshot Array (**1146**)

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7. Hard Hashing Problems

(For top companies)

1. Longest Consecutive Sequence (**128**)
2. Minimum Window Substring (**76**)
3. Substring with Concatenation of All Words (**30**)
4. Alien Dictionary (**269**)
5. Word Ladder (**127**)

Must-Do Interview Hashing Sheet (Top 20)

Solve these first:

1. Two Sum (1)
2. Contains Duplicate (217)
3. Valid Anagram (242)
4. Group Anagrams (49)
5. Top K Frequent Elements (347)
6. Longest Consecutive Sequence (128)
7. Subarray Sum Equals K (560)
8. Longest Substring Without Repeating Characters (3)
9. Minimum Window Substring (76)
10. Find All Anagrams in a String (438)
11. Majority Element (169)
12. Isomorphic Strings (205)
13. Word Pattern (290)
14. Happy Number (202)
15. Continuous Subarray Sum (523)
16. Contiguous Array (525)
17. Time Based Key Value Store (981)
18. LRU Cache (146)

19. Insert Delete GetRandom O(1) (380)
20. Design HashMap (706)

Pattern Recognition (Very Important)

Use HashSet when:

Keywords:

- duplicate
- unique
- exists
- seen before
- cycle detection

Example:

    contains duplicate

Use HashMap when:

Keywords:

- frequency
- count
- mapping
- pair lookup
- index lookup

Example:

    find target sum pair

Use Prefix Sum + HashMap when:

Keywords:

- subarray sum
- count subarrays
- longest subarray

Example:

    subarray sum equals k

Use HashMap + Sliding Window

Keywords:

- substring
- longest
- minimum window
- repeating characters

Example:

    longest unique substring

For your interview preparation I recommend this order:

Basic Hashing → HashSet → HashMap → Prefix Sum + HashMap → String Hashing → Sliding Window + HashMap → Design Questions (LRU, HashMap)